



**Politecnico
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Filtering, Distillation, and Hard Negatives for Vision-Language Pre-Training

Original Authors

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INTRODUCTION AND CAT METHOD

Background: CLIP vs DiHT

CLIP established the foundation for modern multimodal learning.



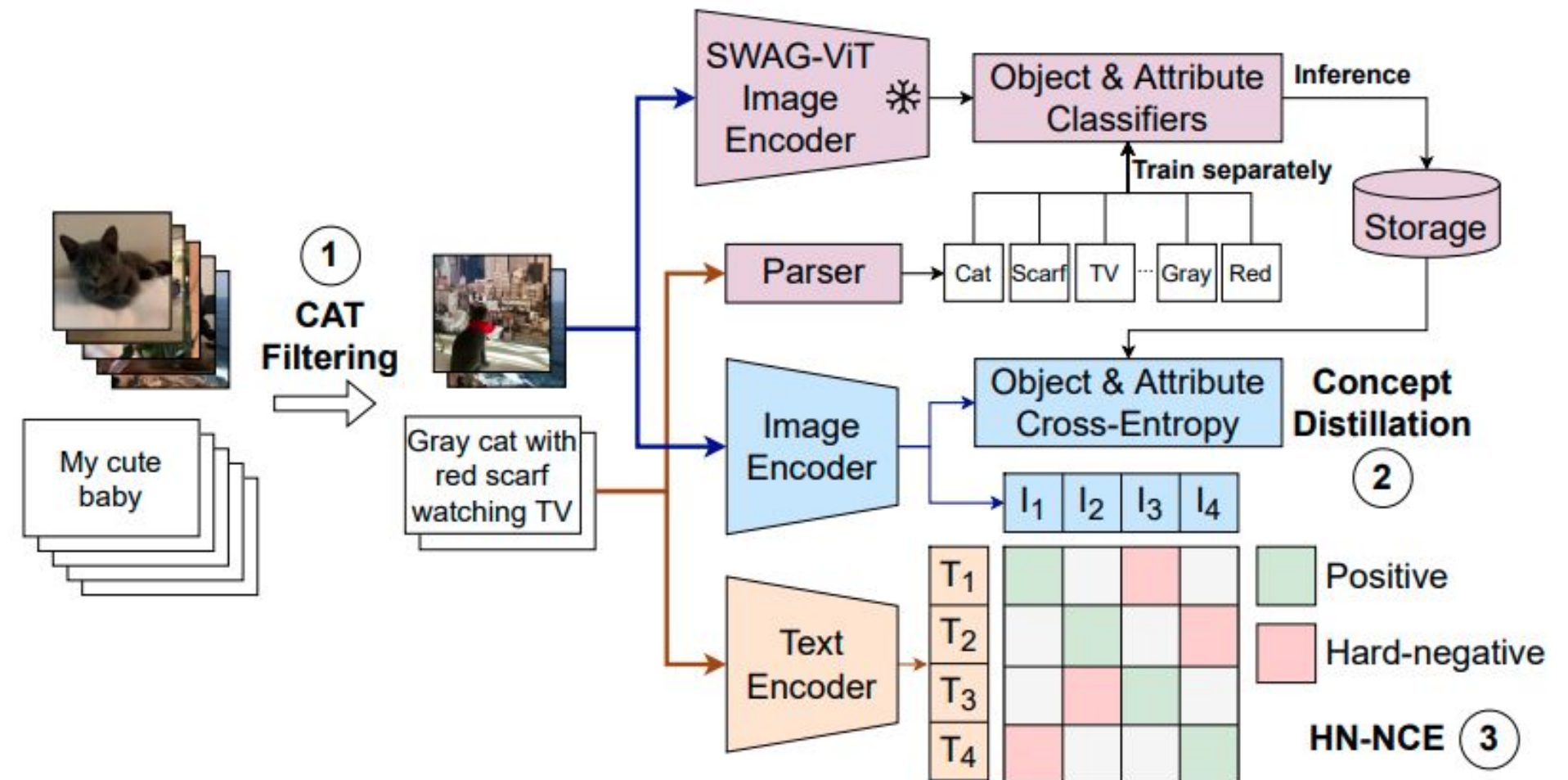
Shared Embedding Space

Projects images and text into a shared geometric space, allowing to make comparisons.



Contrastive Learning

Aligns relevant image-text pairs while distancing unrelated content.



PROPOSED MULTIMODAL PIPELINE DIAGRAM

Clip: Limitations

Multimodal learning relies heavily on contrastive pre-training using large-scale, noisy data. While powerful for zero-shot tasks, the pipeline faces critical structural problems:



Dataset Noise

Addressed by: CAT



Model Initialization

Addressed by: CD



Loss Limitations

Addressed by: HN

CAT Filtering: Complexity & Action

⚠ The Problem

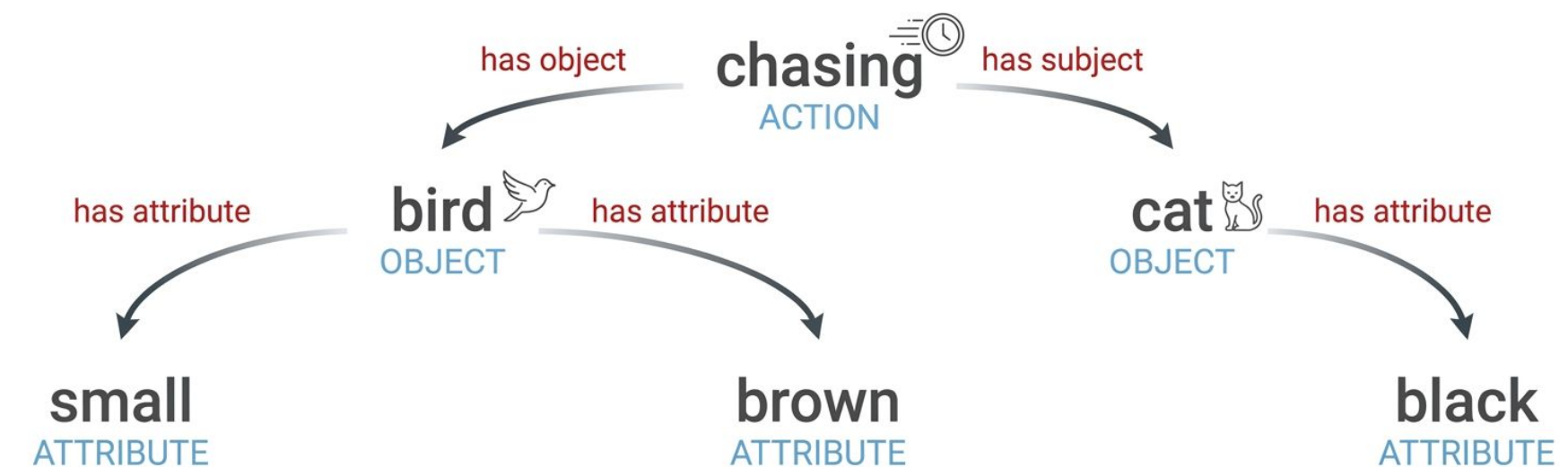
Web-scale datasets contain **noise**: generic labels, poor captions...

💡 The Solution

Filtering by **Complexity Score**. We only maintain pairs with rich semantic relationships (>C1, One action).



Caption: *A black cat is chasing a small brown bird.*





CAT Filtering: Text Spotting

⚠ The Problem

Up to ~30% of web images contain text that matches the caption. The model learns to "read" rather than understand visual-semantic alignment.

✂ The Solution

We apply a filter using a text-spotter:

- Confidence: Score ≥ 0.8
- Match: ≥ 5 characters match the caption



Empirical Results

The core idea: **prioritize** data quality



Performance Gain



Data Efficiency



Subset Optimization

Outperforming standard CLIP
baselines using a training subset
reduced by 80%.



Distillation

Warm-Start Limitations

The Initialization Dilemma

Feature Degradation

Fine-tuning degrades pre-trained features, leading to **catastrophic forgetting** of robust general knowledge.

Encoder Rigidity

Locked encoders fail to adapt to complex queries, limiting the model's expressive capacity.

Retrieval Drop

Cross-modal retrieval **performance drops** significantly when alignment is not properly maintained.



Full Fine-Tuning
High cost, feature drift

Locked-Image Tuning
Inflexible, lower performance

Concept Distillation (CD)

Bridging the Alignment Gap

Teacher Module

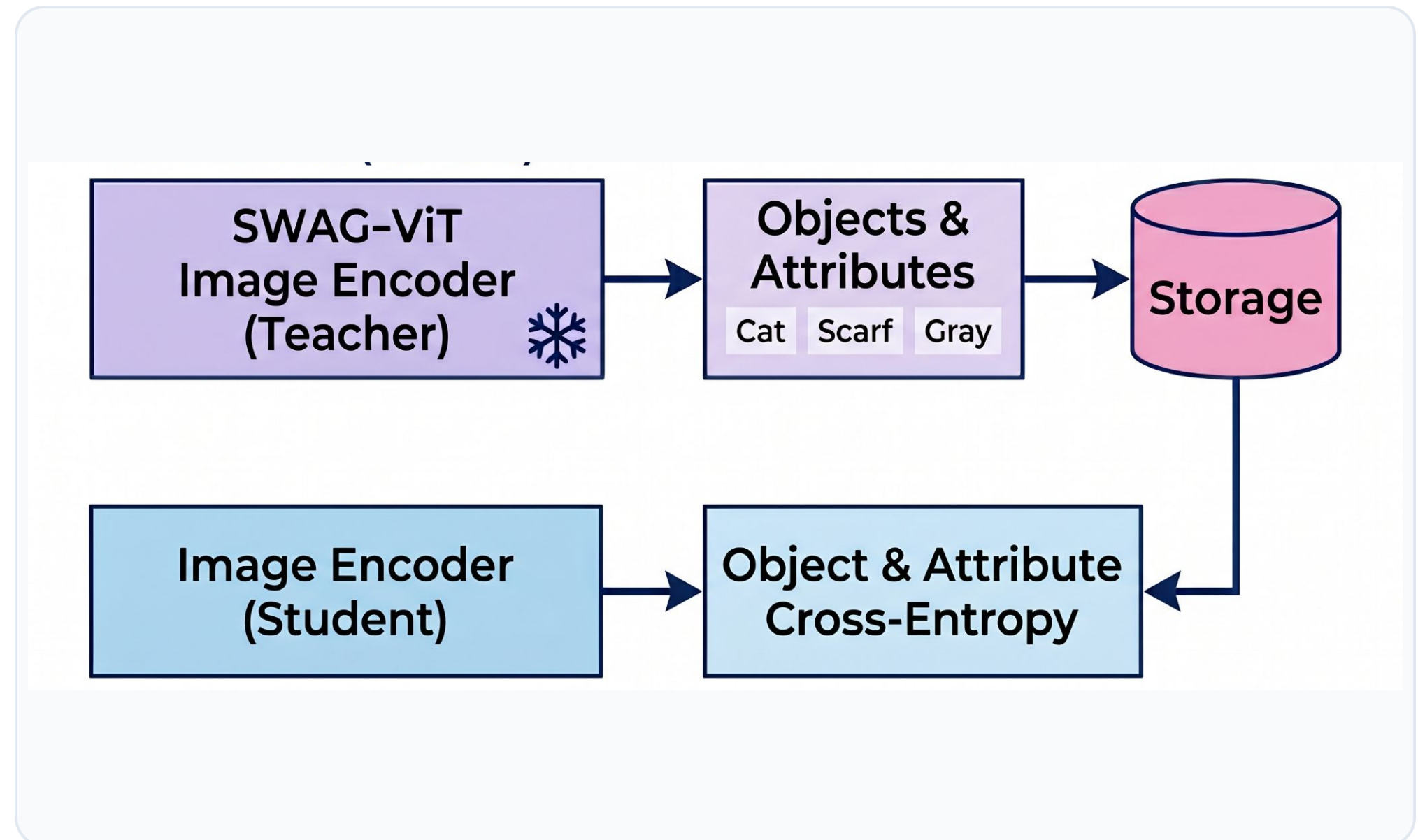
Predicts objects and attributes to guide the student's learning process.

Student Learning

Trains via linear classifiers to reconstruct teacher concepts.

Joint Optimization

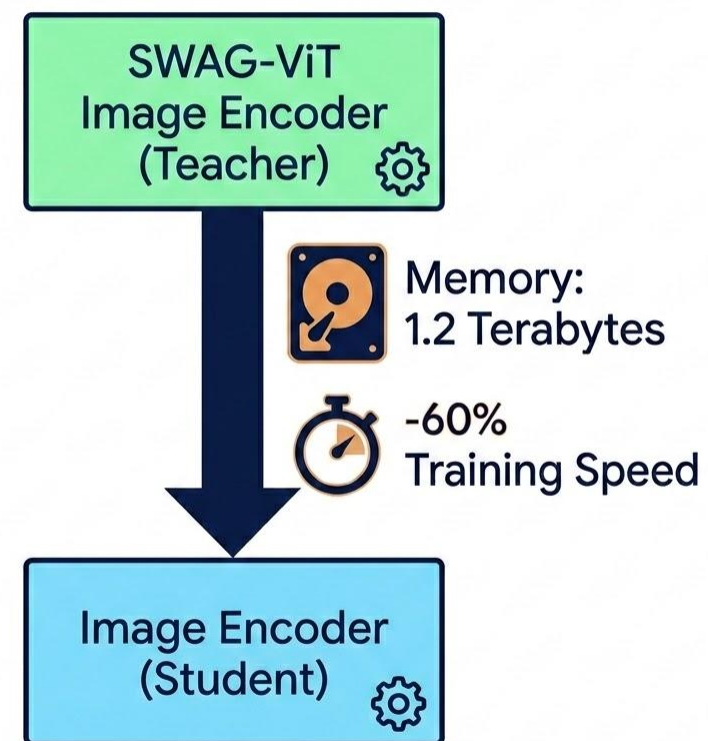
Trained jointly with contrastive loss for superior cross-modal alignment.



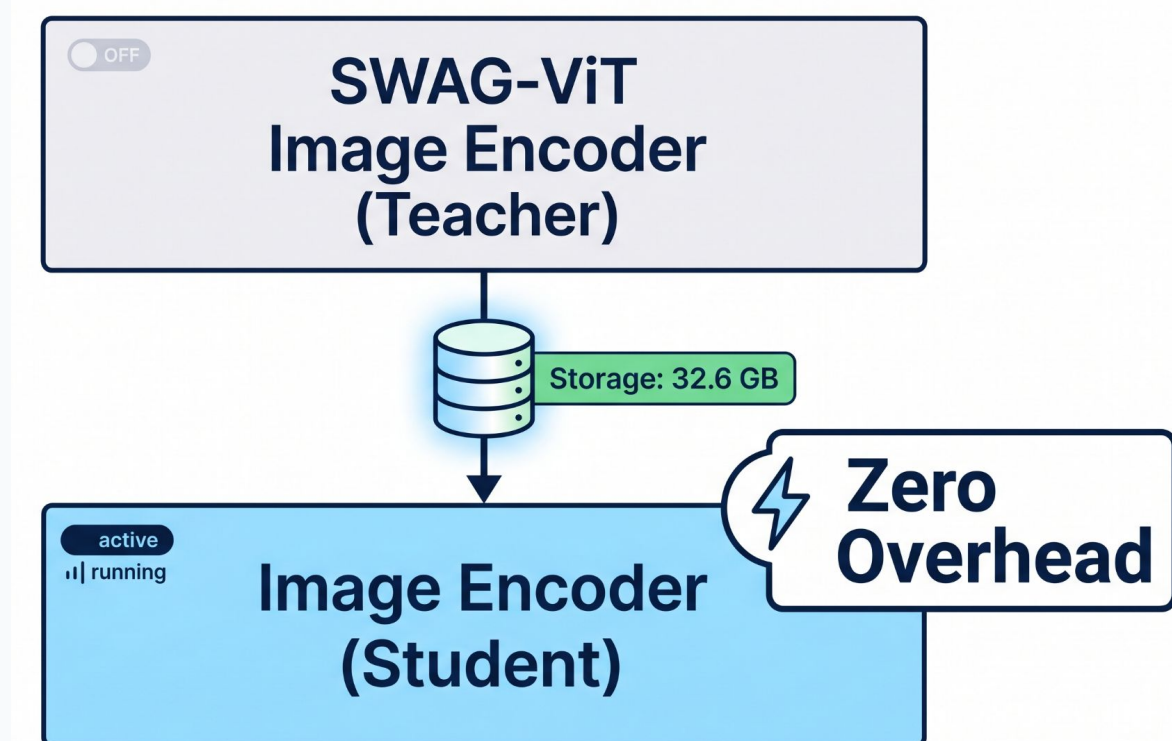
Efficiency & Memory

Zero-Overhead Distillation

Standard Distillation



Concept Distillation



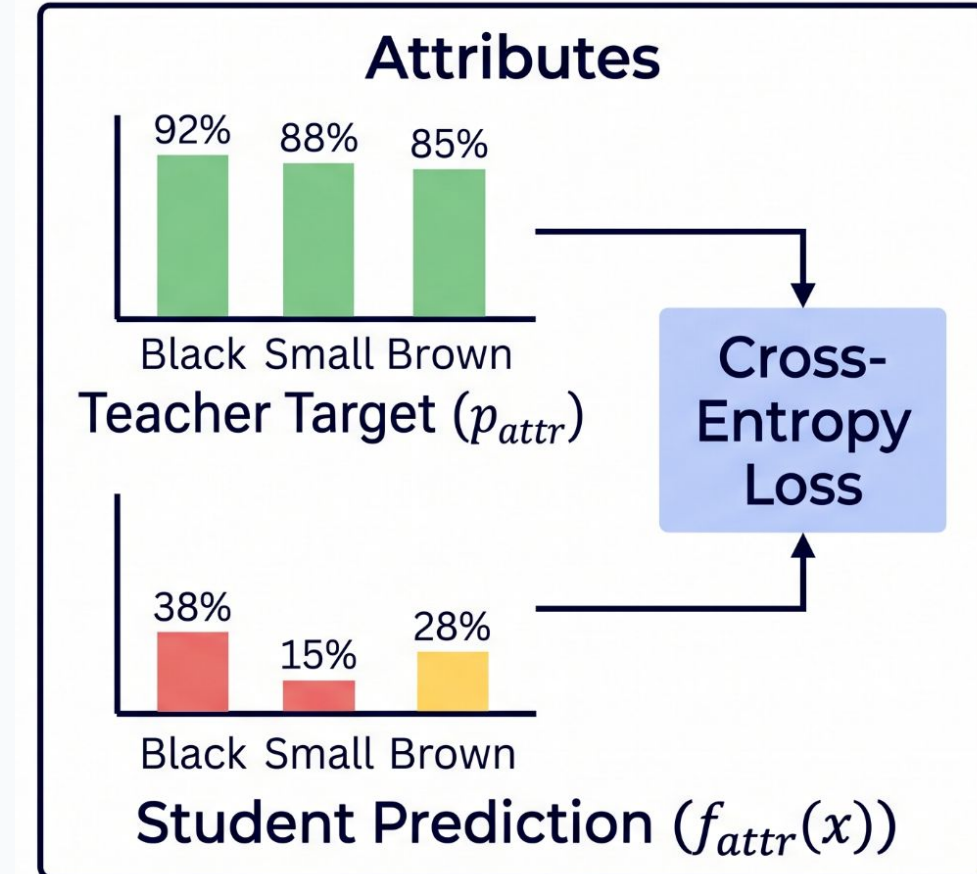
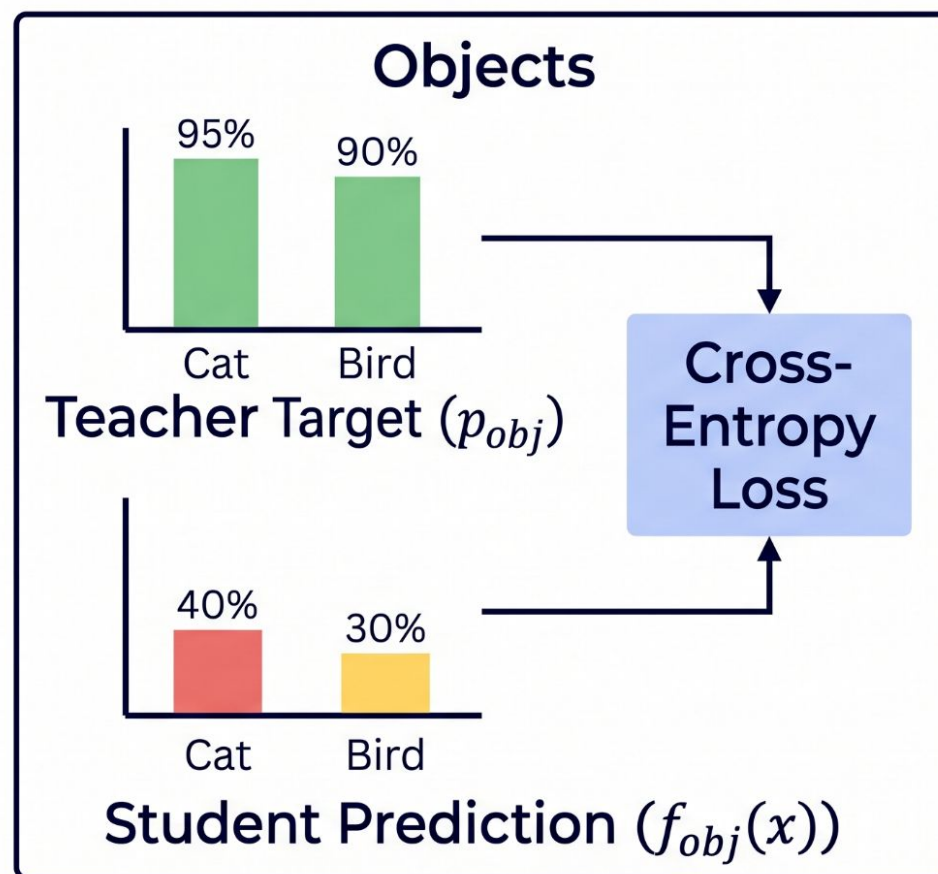
- Teacher predicts objects and attributes.
- Student trains via linear classifiers.

- Trained jointly with contrastive loss.
- Zero-overhead metadata storage.

Alignment via Cross-Entropy

Minimizing the error on objects and attributes predictions"

Sentence: “A black cat is chasing a small brown bird”



$$\mathcal{L}_{\text{CE-O}}(\mathbf{X}) = \sum_{i=1}^n \text{CROSS-ENTROPY}(\mathbf{p}_i^{\text{obj}}, f_{\text{obj}}(\mathbf{x}_i))$$

$$\mathcal{L}_{\text{CE-A}}(\mathbf{X}) = \sum_{i=1}^n \text{CROSS-ENTROPY}(\mathbf{p}_i^{\text{attr}}, f_{\text{attr}}(\mathbf{x}_i))$$

Advantages of Concept Distillation

Why DiHT outperforms Standard Distillation



Richer Labels

Teacher predictions are more exhaustive than noisy web captions.



Zero Overhead

Discarding the Teacher after inference removes memory and compute limits.



Superior to Embedding

Outperforms standard vector distillation.



End of the Dilemma

Enables text alignment flexibility while preserving pre-trained visual features.

Ablation Results

Superiority of Concept Distillation

Init	Method	SWAG (teacher)	IN	COCO		Flickr	
				T2I	I2T	T2I	I2T
Random	Baseline	—	68.7	42.8	60.5	72.8	89.7
	ED	B/16	69.2	42.6	59.4	72.8	86.8
	DD	B/16	68.6	41.8	57.4	71.7	87.0
	CD (ours)	B/16	71.0	42.8	59.5	72.3	86.5
	CD (ours)	H/14	72.3	43.4	60.4	73.8	87.6
SWAG	LiT	—	73.0	32.5	50.6	60.8	79.6
	FT	—	71.2	43.1	60.3	73.1	87.7
	FT-delay	—	72.0	42.7	60.7	72.5	86.2

Setup & Legend:

- **Model:** ViT-B/16 on LAION-CAT
- **ED / DD:** Standard Distillation (Embedding/Distribution)
- **LiT:** Locked-image Tuning (locked vision)
- **FT:** Standard Fine-Tuning
- **CD (Ours):** Concept Distillation

- Outperforms Locked-Image Tuning (LiT).
- Surpasses standard Fine-Tuning (FT).
- Improves both vision and alignment.



Multimodal alignment with hard negatives

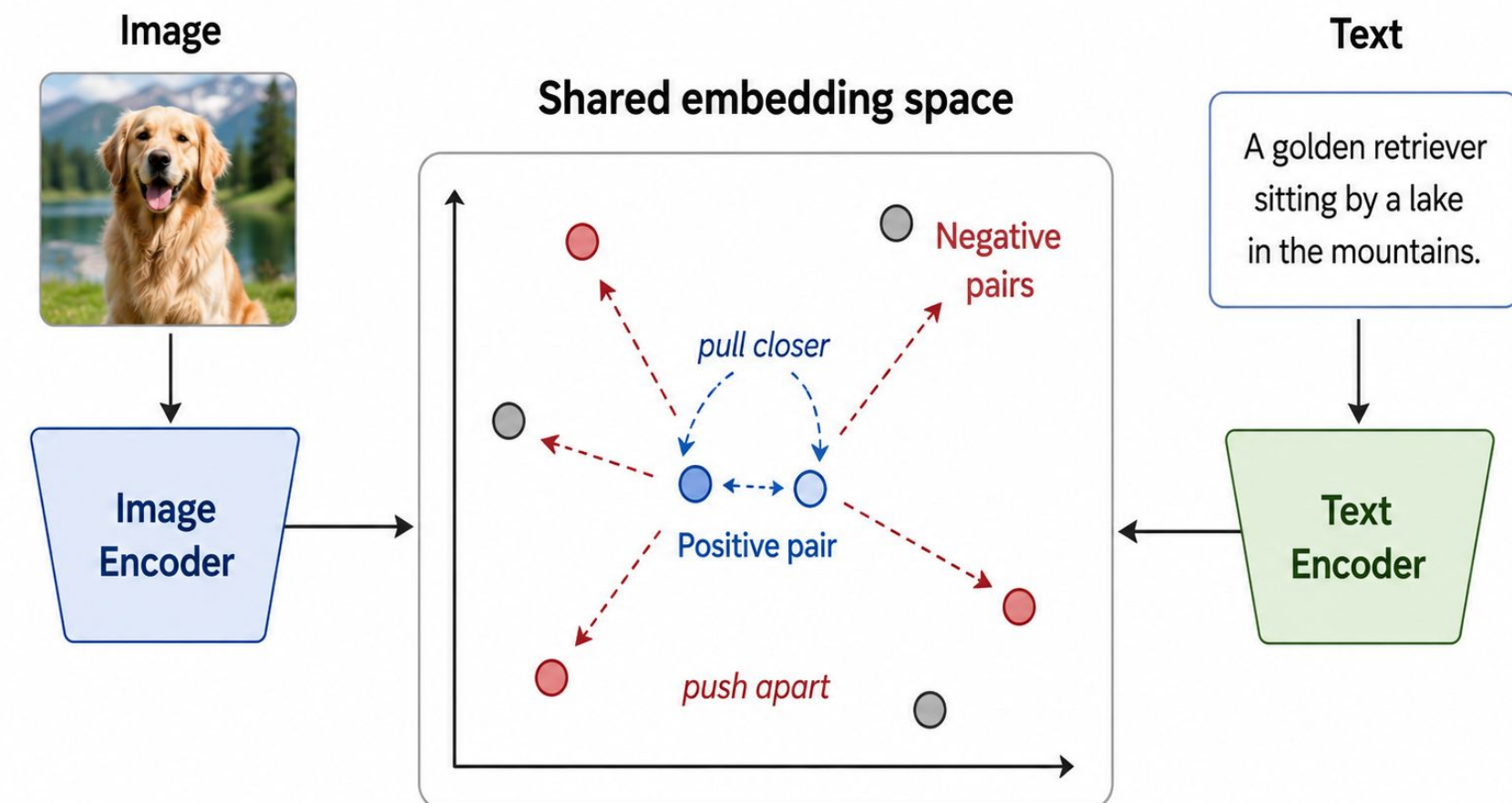
Vanilla InfoNCE: Contrastive Learning Baseline



The de facto approach for image-text alignment.

Vanilla InfoNCE Objective

Pulls positive image-text pairs together and pushes negative pairs apart.



Limitation: all in-batch negatives are treated similarly.

Not all negatives are equally useful

Query



Captions

'A dog running on grass'



Positive pair

'A pizza on a table'



Easy negative,
low similarity

'A dog sitting on grass'



Hard negative,
high similarity

'A brown dog outdoors'



False negative,
semantically valid but treated as
negative



Key idea: Reweights negative samples using **Importance Sampling**.

HN-NCE: Hard Negatives Contrastive Loss

The contrastive alignment component of the final training objective

$$\mathcal{L}_{\text{HN-NCE}}(\mathbf{X}) = -\sum_{i=1}^n \log \left[\frac{e^{\mathbf{x}_i^\top \mathbf{t}_i / \tau}}{\alpha \cdot e^{\mathbf{x}_i^\top \mathbf{t}_i / \tau} + \sum_{j \neq i} e^{\mathbf{x}_i^\top \mathbf{t}_j / \tau} w_{\mathbf{x}_i, \mathbf{t}_j}^{i \rightarrow t}} \right] - \sum_{i=1}^n \log \left[\frac{e^{\mathbf{x}_i^\top \mathbf{t}_i / \tau}}{\alpha \cdot e^{\mathbf{x}_i^\top \mathbf{t}_i / \tau} + \sum_{j \neq i} e^{\mathbf{x}_j^\top \mathbf{t}_i / \tau} w_{\mathbf{x}_j, \mathbf{t}_i}^{t \rightarrow i}} \right].$$

$$w_{\mathbf{x}_i, \mathbf{t}_j}^{i \rightarrow t} = \frac{(n-1) \cdot e^{\beta \mathbf{x}_i^\top \mathbf{t}_j / \tau}}{\sum_{k \neq i} e^{\beta \mathbf{x}_i^\top \mathbf{t}_k / \tau}}$$

$$w_{\mathbf{x}_j, \mathbf{t}_i}^{t \rightarrow i} = \frac{(n-1) \cdot e^{\beta \mathbf{x}_j^\top \mathbf{t}_i / \tau}}{\sum_{k \neq i} e^{\beta \mathbf{x}_k^\top \mathbf{t}_i / \tau}}$$

- Reweights negatives according to their image-text similarity.
- β controls the strength of hard-negative reweighting in the denominator.
- α adjusts the positive term to account for false negatives in the denominator.

Final Training Objective

$$\mathcal{L}_{\text{HN-NCE}}(\mathbf{X}) + \mathcal{L}_{\text{CE-O}}(\mathbf{X}) + \mathcal{L}_{\text{CE-A}}(\mathbf{X})$$

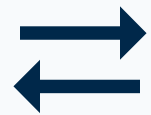


Image-Text Alignment



Object Concept Knowledge



Attribute Concept Knowledge

Effect of Hard Negative Contrastive Training

#	Method		IN	COCO		Flickr	
	CD	HN		T2I	I2T	T2I	I2T
1			68.7	42.8	60.5	72.8	89.7
2	✓		72.3	43.4	60.4	73.8	87.6
3	✓	✓	72.0	43.7	62.0	73.2	89.5

CD: Concept Distillation

HN: Hard Negatives

Setup: ViT-B/16 trained on LAION-CAT for 16B processed samples.

- HN-NCE improves retrieval over CD alone
- Stronger gains on image-to-text retrieval
- Effects are complementary to Concept Distillation



Experiments

Experiments Setup

Architecture

- **Encoders:** ViT Vision Encoder (B/32, B/16, L/14) & Text Transformer.
- **Optimizer:** Adam with decoupled weight decay & mixed-precision.
- **Scheduler:** Cosine Learning Rate
- **Hyperparameters:** grid-search on B/32

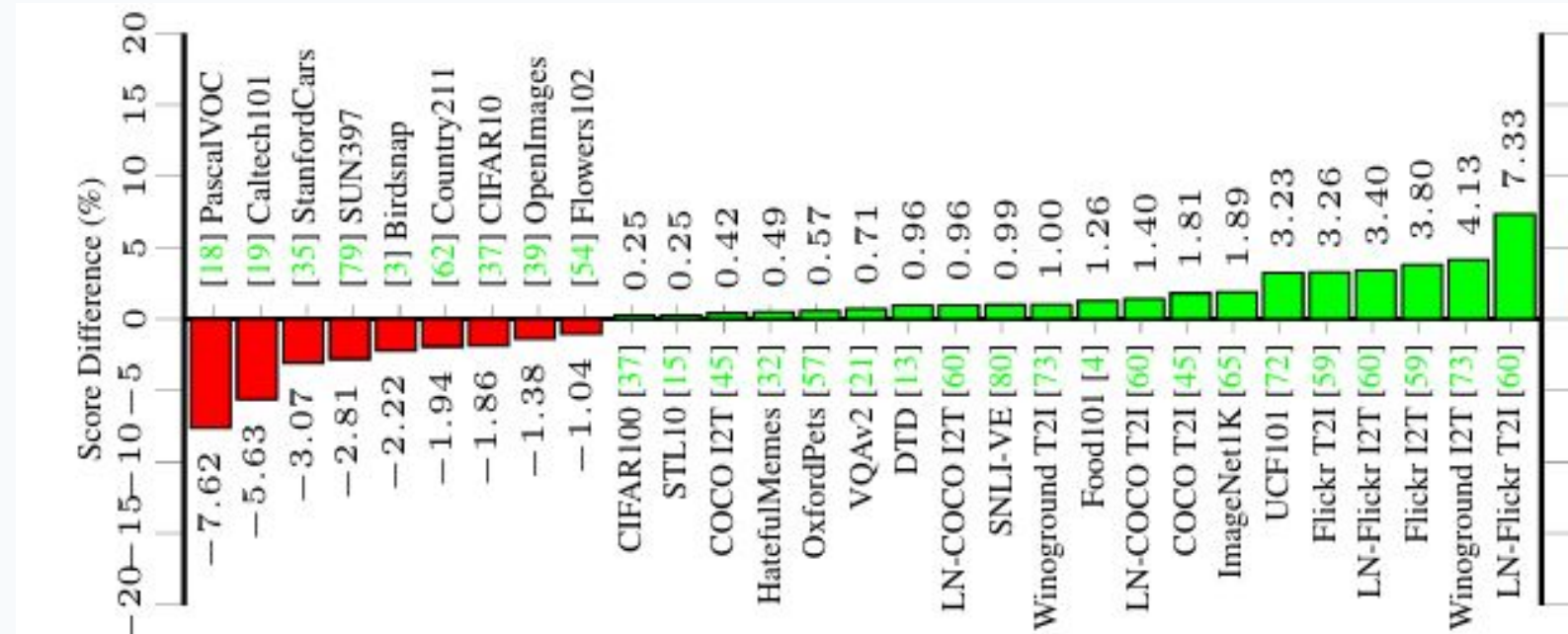
Training Datasets

- 2.1B image-caption pairs subset of the **LAION-5B**
 - NSFW content removal
 - Watermark filtering (>50% probability)
- **LAION-CAT:** 22% of the original dataset (480M)
- **PMD:** 63M of public datasets.

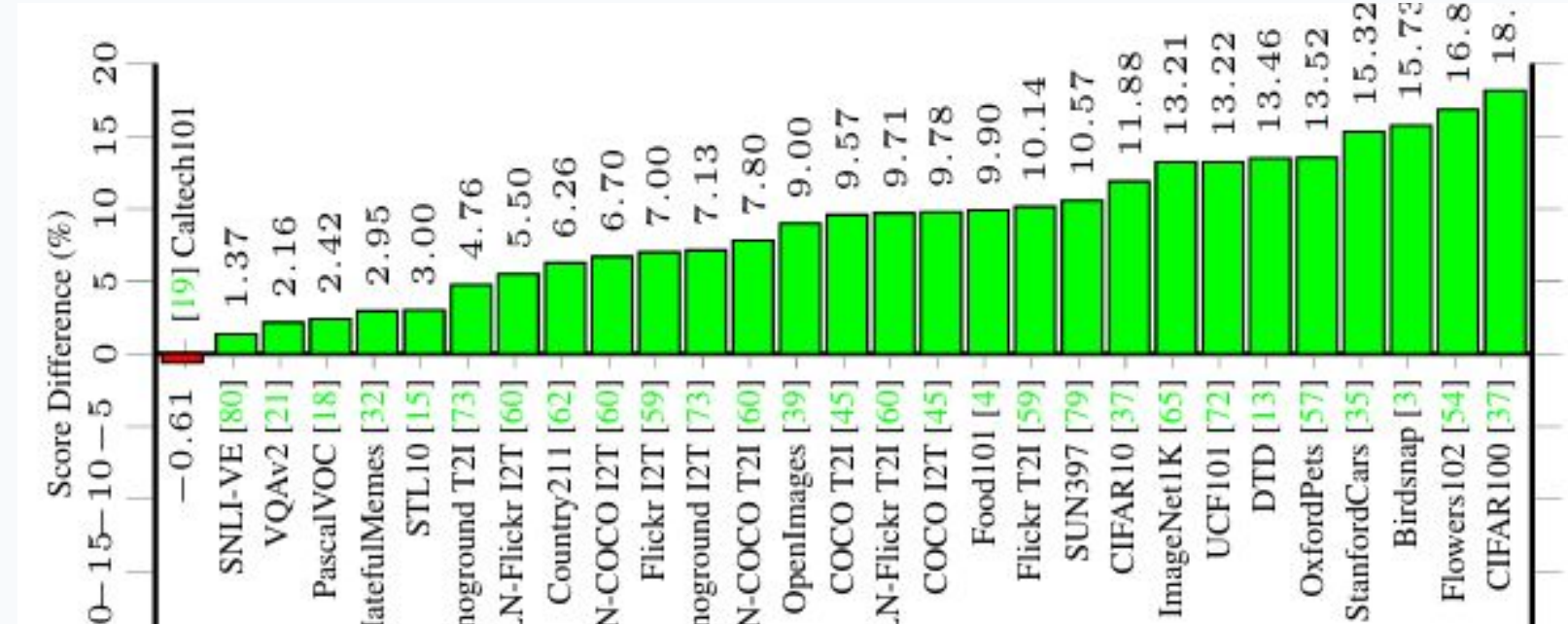
Evaluation Metrics

- Zero-Shot Classification
- Few-shot Linear Probing
- Robustness to Distribution Shifts

Zero-Shot Benchmark Results



DiHT-B/16 trained on LAION-CAT vs. CLIP-B/16 trained on LAION-2B



DiHT-B/16 vs. CLIP-B/16

Both trained on PMD.

Zero-Shot Benchmark

- **Image Classification:** 17 tasks (incl. ImageNet-1K).
- **Cross-Modal Retrieval:** 10 tasks (COCO & Flickr T2I/I2T).
- **Visual Question Answering:** 2 tasks.

State of Arts & Robustness

Method	px	#P	#D	#S	IN	COCO		Flickr		
						T2I	I2T	T2I	I2T	
ViT-B/32										
CLIP [62]	224	151M	400M	12.8B	63.4	31.4	49.0	59.5	79.9	
OpenCLIP [27]	224	151M	400M	12.8B	62.9	34.8	52.3	61.7	79.2	
OpenCLIP [27]	224	151M	2.3B	34B	66.6	39.0	56.7	65.7	81.7	
DiHT	224	151M	438M	16B	67.5	40.3	56.3	67.9	83.8	
DiHT	224	151M	438M	32B	68.0	40.6	59.3	68.6	84.4	
ViT-B/16										
CLIP [62]	224	150M	400M	12.8B	68.4	33.7	51.3	63.3	81.9	
OpenCLIP [27]	224	150M	400M	12.8B	67.1	37.8	55.4	65.2	84.1	
OpenCLIP [27]	240	150M	400M	12.8B	69.2	40.5	57.8	67.7	85.3	
DiHT	224	150M	438M	16B	71.9	43.7	62.0	73.2	89.5	
DiHT	224	150M	438M	32B	72.2	43.3	60.3	72.9	89.8	
ViT-L/14										
CLIP [62]	224	428M	400M	12.8B	75.6	36.5	54.9	66.1	84.5	
CLIP [62]	336	428M	400M	13.2B	76.6	37.7	57.1	68.6	86.6	
OpenCLIP [27]	224	428M	400M	12.8B	72.8	42.1	60.1	70.4	86.8	
OpenCLIP [27]	224	428M	2.3B	32B	75.2	46.2	64.3	75.4	90.4	
DiHT	224	428M	438M	16B	77.0	48.0	65.1	76.7	92.0	
DiHT	336	428M	438M	16.4B	77.9	49.3	65.3	78.2	91.1	

Legend

- **px**: input size
- **#P**: model size
- **#D**: training dataset size
- **#S**: total sample processed

ImageNet robustness performance ViT-B/16 models

Method	Train Data	#D	$\mathbb{Z}$	IN-V2	IN-A	IN-R	IN-Sketch	ObjectNet
CLIP	PMD	63M	54.6	47.9	35.5	56.3	30.8	38.5
DiHT	PMD	63M	67.8	61.5	54.3	74.4	44.7	52.8
CLIP	LAION-2B	1.98B	70.3	62.7	39.3	81.0	57.1	56.2
DiHT	LAION-CAT	438M	72.2	64.3	49.2	85.1	58.3	62.3



Few-Shot Part 1

Best Scenario & Problem

Warm Start



Pre-training on a large, high-diversity base dataset to learn robust features.

Feature Re-use



Extracting embeddings that capture universal patterns (edges, shapes, semantic cues).

Few-Shot Task



Fine-tuning or projecting new samples into the learned space for instant classification.

Theory vs Practice

Perform extremely poorly on low-data regime

Few-shot - Part 2

Best Scenario & Problem

Methodology

- **Initialization:** Classifier weights are calibrated using zero-shot text prompts as informative priors for each class.
- **Constraint:** Projected Gradient Descent (PGD) is applied to ensure weights do not drift excessively from initial prompts.

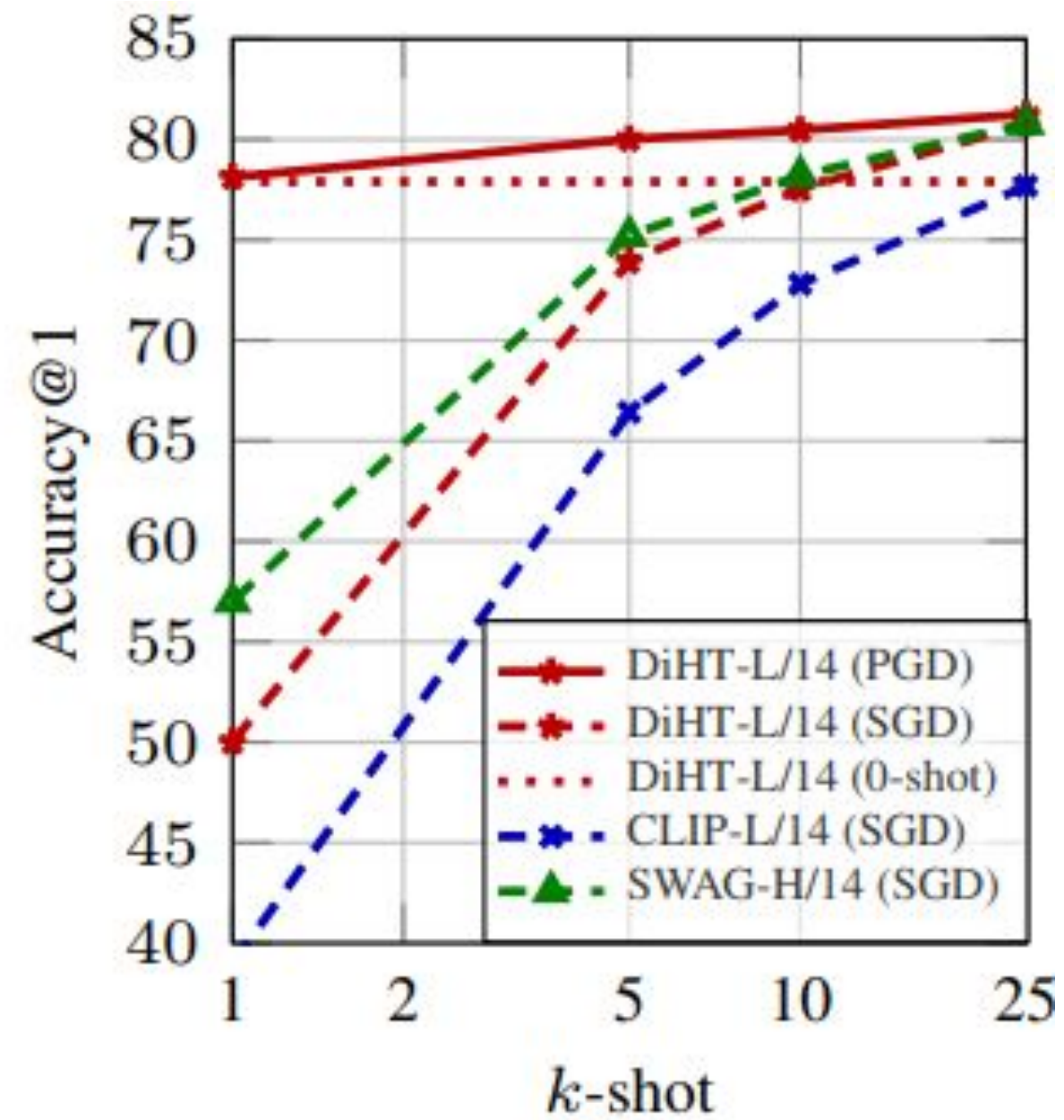
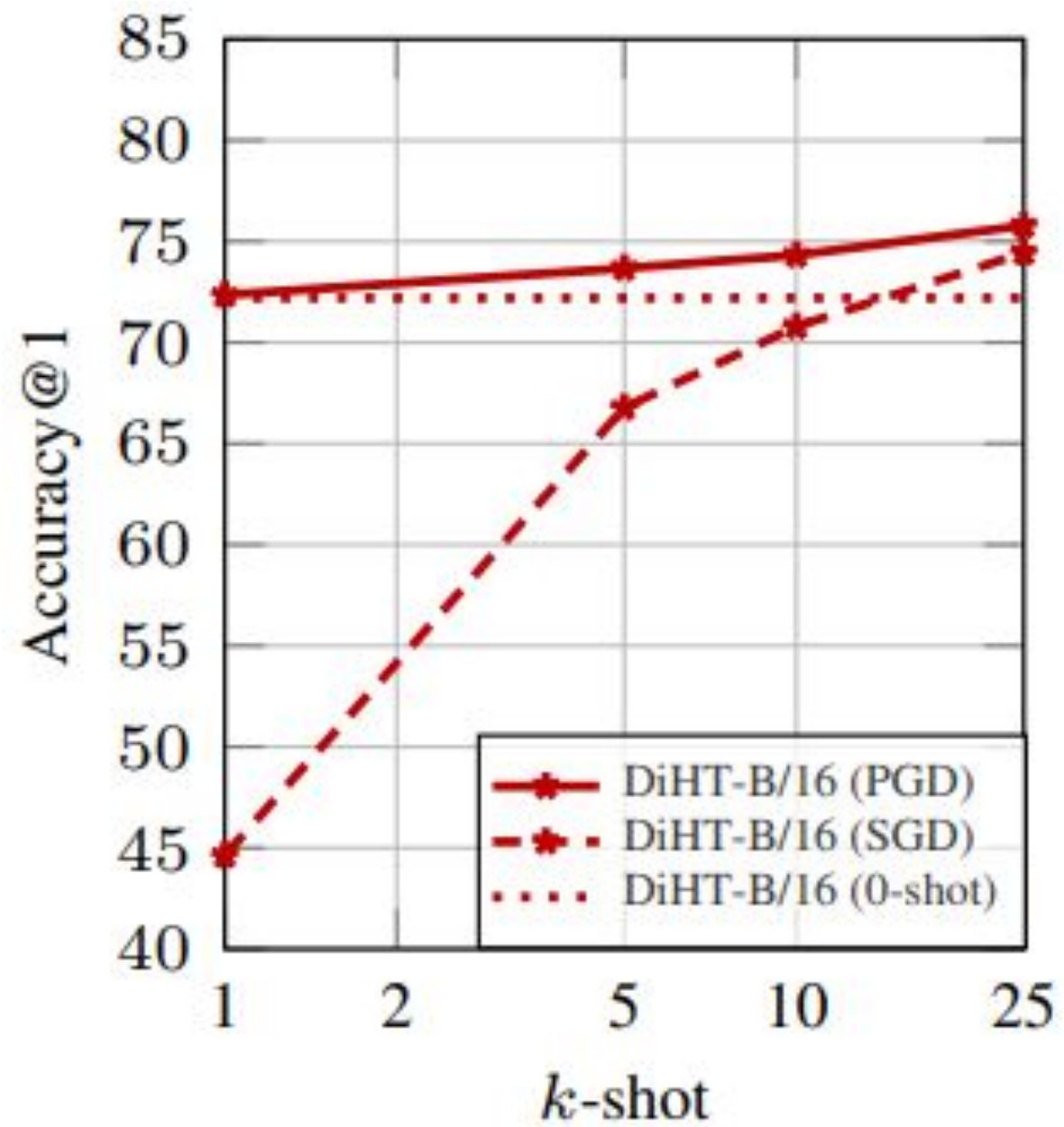
$$\min_{\|\mathbf{W}\|_2 \leq \delta, \|\mathbf{b}\|_2 \leq \delta_b} \sum_{i=1}^n \mathcal{L}_{\text{CE}} \left(y_i, \mathbf{x}_i^\top (\mathbf{W} + \mathbf{W}_0) + \mathbf{b} \right)$$

Hyper Parameters

prompt initialization

biases

Few-Shot Part 3



At first glance

- Performs better with small K
- With larger K it is close to the other models

Conclusions & Future Works

Conclusions

CAT Filtering

Can be applied on every large-scale dataset for improved performance with smaller training schedules

Concept Distillation

Presents a compute and storage efficient way of leveraging very large capacity pre-trained image models for multimodal training

Hard-negatives loss

The benefits are less on noisier LAION dataset compared to PMD

Projected Gradient Descent

Covers the crucial performance gap between zero-shot and few-shot learning

Future Work

New Architectures

Multi-modal encoder-decoder architectures that are more expensive but have zero-shot performance compared to dual encoders.



**Thank you
for your
attention!**